

❖ **Determine the composition and temperature of the exhaust products of a constant-pressure adiabatic combustion process.**

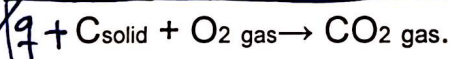
(a) chemical transformation and. (b) composition of products.

$$\alpha A + \beta B + \dots \xrightarrow{\text{c.t.e.}} \mu M + \nu N + \dots$$
 reactant products

chemical Equ. means that
there is no change in concentration
of reactants & products

The constant- pressure heat of formation of a substance is the heat interaction that occurs when one mole of the substance is formed at constant pressure and temperature from its elements as they occur in nature.

- Heat of formation of carbon dioxide (-393.522 kJ/gmol) at 298 K:



endothermic

- System must give off heat to maintain (or return to) the reference temperature.

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endothermic (reactions that absorb heat)

exothermic (" " release ") $\text{CH}_4 + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$

Stoichiometric ratio: it refers to the exact air-to-fuel ratio needed for complete combustion. At this ratio, no excess fuel or O_2 leaves at products.

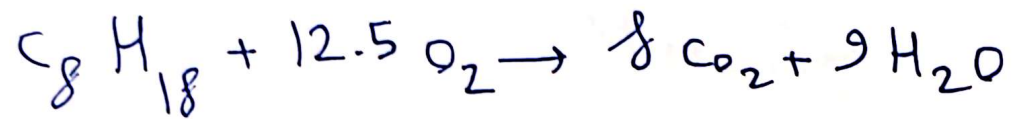
i.e., no unburnt fuel or residual O_2



reactants

products

Verify: Compute stoichiometric ratio for this reaction ^{with HW#5}



if we have more than 12.5 moles O_2 , ^{and 1 mole of octane} our reaction is called air-rich

if we have more than 1 mole of octane & 12.5 moles of O_2 , reaction is called fuel-rich

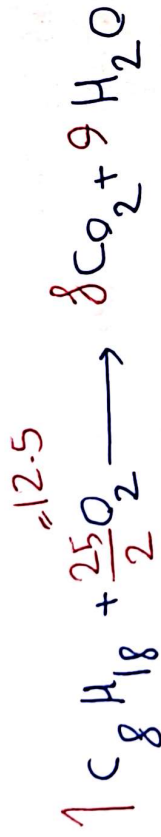
air-rich $\rightarrow \frac{S}{S_{st}} > 1$ (More than 12.5 moles O_2
for 1 mole ~~the~~ octane)
fuel rich $\rightarrow \frac{S}{S_{st}} < 1$ $\rightarrow$ (More than 1 mole
octane for 12.5 moles
 O_2)

Example: gasoline reaction with O_2 , Compute stoichiometric ratio

can be approximated with octane, i.e. C_8H_{18}

$$S = \frac{\text{mass of air}}{\text{mass of fuel}}$$

complete reaction



12.5 moles of O_2 are required to completely burn 1 mole of octane.

$$\text{Mass of octane} : 1 \text{ mole} \times [8 \times 12 + 18 \times 1] \frac{g}{\text{mole}} = 114 g$$

$$O_2 : 2 \times 16 \times 12.5 = 400 g$$

O_2 is 23% of air, so 400g of O_2 is in 1739.13g of air

$$S = \frac{\text{Mass of air}}{\text{mass of fuel}} = \frac{1739.13 g}{114 g} = 15.25$$